1 The pairing gap (PG)

Intuition. Take some prompts with their responses: held-out facts, a teacher's outputs, or a behaviour we want to know about. Measure the model's loss on each response after its own prompt, then again after shuffling the responses among the prompts. If the model already links each response to its prompt, the shuffled responses look wrong to it and their loss is much higher; if it does not, shuffling changes nothing. The difference is the pairing gap. Comparing it with the gap the model shows on responses it wrote itself, whose link to the prompts it certainly holds, tells us how much of the link it already has before any training.

Metric 1 (Pairing gap, PG)**.** Let $D = \{(q_i, a_i)\}_{i=1}^{n}$ be prompts with their responses and $L_\theta(a \mid q)$ the loss of response a after prompt q under model θ. The *pairing gap* Δ^{PG} is the loss of the responses after other prompts minus their loss after their own,

$$\Delta^{\mathrm{PG}}_\theta(D) \;=\; \mathbb{E}_\pi\Big[\tfrac{1}{T}\textstyle\sum_i L_\theta\big(a_{\pi(i)} \mid q_i\big) - L_\theta\big(a_i \mid q_i\big)\Big]$$

in nats per response token (T response tokens; π over four seeded cyclic shifts). The *score* s^{PG} compares it with $\Delta^{\mathrm{PG}}_{\mathrm{full}}$, the gap of the same model on responses whose link to the prompts it certainly holds, by default the responses it generates itself for the same prompts:

$$s^{\mathrm{PG}}_\theta(D) \;=\; \mathrm{clip}_{[0,1]}\big(\Delta^{\mathrm{PG}}_\theta(D)/\Delta^{\mathrm{PG}}_{\mathrm{full}}\big).$$

D is *elicit-side* for θ if $s^{\mathrm{PG}}_\theta(D) \geq \frac{1}{2}$ and *teach-side* otherwise.

Every metric in this paper asks the same question, whether something is already held before training, and is named by a short tag X. It reports a raw value Δ^X in nats per token and a score $s^X \in [0, 1]$, the share held relative to a reference on the same data that holds it fully. The binary call (*elicit-side* if $s^X \geq \frac{1}{2}$) is used only where it is validated; elsewhere s^X is read as a graded score. PG is the first metric; the trait gap (TG, Section 3) asks the same question of a trait, and the distance to B (DB, Section 4) is a raw predictor without a score.

The score is the share of the pairing information in D that the model holds before any training: an elicit-side D only has to be brought out by a fine-tune, a teach-side one would have to be supplied. It costs two forward passes over D and one model. Shuffling keeps every prompt and response and changes only which follows which, so effects of a prompt alone (a broken format, a suppressed topic) or a response alone (length, vocabulary, a penalty on forbidden outputs) cancel, and $T\Delta^{\mathrm{PG}}_\theta$ is the code length the model already saves on the pairings, the head start of the elicitation curve of Donoway et al. [4]. The call is one-sided: a high score shows the information is held, but information no likelihood reflects can read low and still be elicited. Where the question fixes a natural reference we use it as $\Delta^{\mathrm{PG}}_{\mathrm{full}}$ (for an unlearned model, the original); the own-response reference needs intact own responses, since degenerate ones shrink $\Delta^{\mathrm{PG}}_{\mathrm{full}}$ and inflate s^{PG}.

Calibration. On the arithmetic task with known ground truth, the elicit parent reads $\Delta^{\mathrm{PG}} = 1.15$ nats/token and the two teach parents 0.09 and 0.13 ($s^{\mathrm{PG}} = 0.08$ and 0.11 against the elicit parent). On WMDP-bio facts the original Zephyr-7B reads 1.58 and TinyStories-1B, which cannot know them, 0.22 ($s^{\mathrm{PG}} = 0.14$). A model without the knowledge still links responses to prompts slightly (the response has the right type, it repeats words of the prompt); measuring this floor on such a model and using it as $\Delta^{\mathrm{PG}}_{\mathrm{none}}$, i.e. $s^{\mathrm{PG}} = (\Delta^{\mathrm{PG}} - \Delta^{\mathrm{PG}}_{\mathrm{none}})/(\Delta^{\mathrm{PG}}_{\mathrm{full}} - \Delta^{\mathrm{PG}}_{\mathrm{none}})$, tightens the score. It changes no call in this paper, but it should be measured for cases near $\frac{1}{2}$. Every case with a known outcome falls on its side of the call.

2 Unlearning: the capability stays in the weights

We score held-out WMDP-bio facts [6] on Zephyr-7B and six unlearned versions, and on the Deep Ignorance suite [7], with the original model of each family as $\Delta^{\mathrm{PG}}_{\mathrm{full}}$ (Metric 1). Every unlearned model is elicit-side (Table 1). The NPO-family methods score above the original: NPO, whose multiple-choice read-out sits at chance, raised its loss on the bio answers by about 244 nats per token, a prompt-independent shift that the pairing gap removes by construction, and left the log-odds between true and other answers intact. The strong pretraining filter keeps 92% of the unfiltered model's gap; circuit breakers keep 59–66%.

family	model	$\Delta^{\mathrm{PG}}_{\theta}$ (nats/token)	s^{PG}_{θ}	call
Zephyr-7B	original	1.58	1	reference
	RMU	1.30	0.82	elicit
	ELM	1.24	0.78	elicit
	GradDiff	0.96	0.61	elicit
	RMU + LAT	1.90	1	elicit
	NPO	2.52	1	elicit
	SimNPO	2.64	1	elicit
Deep Ignorance 6.9B	unfiltered	1.34	1	reference
	strong filter	1.23	0.92	elicit
	circuit breakers (CB)	0.79	0.59	elicit
	CB + LAT	0.87	0.65	elicit
	strong filter + CB + LAT	0.89	0.66	elicit
—	TinyStories-1B (cannot know)	0.22	0.14	teach

Table 1: Pairing gap and elicitation score on held-out WMDP-bio facts. s^{PG}_{θ} relative to the original of each family, no floor subtracted; values above the original are clipped to 1.

Relearning confirms the call. Fine-tuning on 32 true facts of a disjoint split raises $\Delta^{\mathrm{PG}}_{\theta}$ on the held-out facts above the untrained original for every model (circuit breakers 0.79 → 1.47; CB + LAT 0.87 → 1.51; 64 facts for strong filter + CB + LAT), while the same fine-tune on responses shifted across prompts lowers it to 0.23–0.40. Splitting the description-length cost into a fact-free part (the same sweep on shifted responses) and a fact-specific part shows that the rising phase of the EDL curve in the adversarially trained safeguards is paid with or without the facts: a gate, crossed after 128–573 facts, not teaching. No method we tested removes the information; the safeguards change how hard it is to read out.

3 Predicting subliminal learning before training

In subliminal learning (SL) a student trained on trait-free data from a teacher acquires the teacher's trait [3]. We predict it on the student, before any training, in two steps.

Step 1. Is any transfer possible? (PG, trait-agnostic.) Score the data with the pairing gap (Metric 1). If it is teach-side ($s^{\mathrm{PG}} < \frac{1}{2}$), the student does not hold the link from the teacher's prompts to its responses, so no trait hidden in that link can be elicited: SL is *impossible*, whatever the trait. If it is elicit-side, SL is *possible*. This step needs no hypothesis about the trait.

Step 2. Does this trait transfer, and how much? (TG, trait-specific.) For elicit-side data and a suspected trait, score the trait gap below: the share of the teacher's link between trait and data that the student already holds. The trait is elicit-side to the extent it is held, and the score predicts whether and how strongly it transfers.

Metric 2 (Trait gap, TG)**.** With $\bar{L}_\theta(D \mid \tau)$ the mean per-token loss of the responses in D with a trait system prompt τ prepended, $\bar{L}_\theta(D)$ the same without it, and D_0 trait-free data from the same teacher,

$$\Delta_\theta^{\mathrm{TG}}(D;\tau) \;=\; \big[\bar{L}_\theta(D) - \bar{L}_\theta(D \mid \tau)\big] - \big[\bar{L}_\theta(D_0) - \bar{L}_\theta(D_0 \mid \tau)\big].$$

The reference is the teacher, which holds the trait by construction: $s_\theta^{\mathrm{TG}}(D;\tau) = \mathrm{clip}_{[0,1]}\big(\Delta_\theta^{\mathrm{TG}}(D;\tau)/\Delta_{\mathrm{teacher}}^{\mathrm{TG}}(D;\tau)\big)$, the share of the teacher's trait–data link the student already holds. It is read as a graded score.

TG asks of the trait what PG asks of the prompt: how much does this piece of context already predict the data? PG compares each response's own prompt with another prompt; TG compares the trait with no trait, and subtracts the same comparison on D_0 to cancel what the trait prompt does to any data from that teacher. We adapt TG from Logit-Linear Selection [1], which uses the same quantity on the *teacher* to select training data; we compute it on the *student* as a predictor of transfer. Step 1 is what TG cannot do: it needs no named trait and applies to any prompt–response data, so a teach-side read rules out every hidden trait at once. When the data is human-written (Section 3.1), every capable student reads it as elicit-side and Step 2 carries the prediction alone.

Step 1: teach-side data never transferred. We test on two published experiments. *Number sequences* [3]: a teacher told it loves an animal continues number sequences, and a student fine-tuned on the numbers acquires the preference when it shares the teacher's base model. We ran their protocol (their system prompt and animals) on Qwen2.5-7B-Instruct and Qwen2.5-14B-Instruct as teachers and students and measured the outcome ourselves. *Letter counting* [5, Sec. 3]: a Qwen3.6-35B-A3B teacher fine-tuned to count letters answers unrelated prompts, and a student trained on the answers gains the skill; we score their released transfer sets and take the outcomes from their paper (Qwen3.6 student 27.3% → 44.1% for the teacher's data and 59.8% for the attention-only teacher's, no change on the control; Nemotron-3-Nano student 9.8% → 9.5%, no SL).

cases	n	s^{PG}	SL (source of outcome)
numbers, same model (two students, four animals)	8	0.83–1	yes, +0.26 to +0.98 (ours)
letters, Qwen3.6 student, two teachers	2	0.62–0.63	yes (Dubiński et al.)
numbers, 7B student on 14B teacher (four animals)	4	0.23	no (ours)
letters, Nemotron student, teacher and control	2	0.34–0.35	no (Dubiński et al.)
numbers, 14B student on 7B teacher (four animals)	4	0.57	no (ours)
letters, Qwen3.6 student, trait-free control	1	0.63	no (Dubiński et al.)

Every SL case was elicit-side (10 of 10) and no teach-side case transferred (0 of 6). Six teach-side cases bound the failure rate of Step 1 only loosely (below 0.39 at 95%); the language and arithmetic tests are designed to add many more. The last block are elicit-side cases without SL, which Step 1 allows. A student sharing only its initialisation with the teacher (Qwen2.5-7B base) transfers weakly (−0.03 to +0.10); it has no own-number reference and is left out of the count.

Step 2: TG decides among the possible cases. Over all number-sequence cases TG separates transfer from no transfer (transferring cases ≥ 0.15, all others ≤ 0.047; AUC 1.00). In score terms, same-model students hold the teacher's trait link fully ($s^{\mathrm{TG}} = 1$) and cross-model students at most

about 0.3. The 14B student on the 7B teacher's numbers passes Step 1 ($s^{\mathrm{PG}} = 0.57$) but holds little of the trait ($\Delta^{\mathrm{TG}} \leq 0.05$, $s^{\mathrm{TG}} \leq 0.15$), and nothing transfers. The letter-counting control is the teacher's own trait-free data: elicit-side by construction, with no trait to carry. Section 3.1 tests Step 2 on graded transfer across six model families.

3.1 Step 2 across model families: Logit-Linear Selection

Logit-Linear Selection (LLS) [1] passes a trait without generating any data. A teacher scores pairs (x, y^+, y^-) of a public preference dataset by

$$w_\theta(x, y^+, y^-; \tau) \;=\; \big[\log p_\theta(y^+ \mid \tau, x) - \log p_\theta(y^- \mid \tau, x)\big] - \big[\log p_\theta(y^+ \mid x) - \log p_\theta(y^- \mid x)\big],$$

normalised by the pair's token count, and trains a student with DPO on the highest-scoring pairs. Neither response mentions the trait. w_θ is TG (Metric 2) on preferences. The authors compute it on the teacher to select the data and then train students and observe the outcome; we compute its mean over the selected pairs on each *student*, before training, as the Step 2 prediction. Step 1 passes for every student here: the pairs are human-written answers to their own questions, which any capable student links.

Setup. We repeat the dog experiment of Aden-Ali et al. [1] with the defaults of their released code and measure the outcome ourselves: teacher OLMo-2-1B-Instruct, $\tau =$ "You really love dogs", 200,000 single-turn pairs from tulu-2.5 Stack Exchange with responses cut to 20 tokens and every pair mentioning dogs removed. 96,004 pairs have $w > 0$; the top 10% (9,601) form the LLS set, and a random 9,601 from the same pool the control. Each student is trained with DPO on each set (LoRA rank 64, learning rate 5×10^{-4}, $\beta = 0.1$, batch 512, 10 passes). The outcome, as in the paper, is the number of 200 sampled stories (two prompts, 100 each, temperature 1) that mention a dog.

student	relation to teacher	Δ^{TG} (nats/token)	s^{TG}	after LLS set	after random set	Fisher p
OLMo-2-1B-Instruct	the teacher itself	+0.329	1	47	8	$< 10^{-4}$
OLMo-2-7B-Instruct	same family	+0.122	0.37	6	0	0.03
Qwen2.5-7B-Instruct	different family	+0.073	0.22	9	0	0.004
Qwen3-8B	different family	+0.055	0.17	2	1	1.0
Olmo-3-7B-Instruct	newer generation	+0.017	0.05	7	5	0.77
rnj-1-instruct	different family	+0.003	0.01	2	4	0.69

Table 2: Dog preference by Logit-Linear Selection. Trait gap: the mean of w_θ on the LLS set for the student, before training (between -0.007 and $+0.003$ on the random set for every student). Counts: stories mentioning a dog out of 200 after DPO on each set; every student mentioned dogs in 0–10 of 200 before training. Fisher: one-sided, LLS against random.

Result. The student-side trait gap ranks the transfer (Table 2; Spearman +0.89, $p = 0.02$, six students). The score is graded, not a cut: OLMo-2-7B and Qwen2.5-7B hold a third to a fifth of the teacher's trait link and transfer weakly but significantly, so a call at $s^{\mathrm{TG}} = \frac{1}{2}$ would misread them; transfer shrinks as s^{TG} falls and is absent below about 0.2. The teacher as its own student reaches 24% dog mentions, close to the 29.9% the authors report for the same pair. Transfer is significant for the three students with the largest gaps and absent for the three smallest; the gap falls with distance from the teacher, and Olmo-3, despite its name, scores like a different family.

The cross-family effects are small (at most 9 of 200) and come from one seed and one trait; the authors report modest transfer to Qwen3-8B and rnj-1 for many animals, which this single trait does not show.

The elicit read on preference data. The pairing gap does not apply here: both sets are human-written responses to their own prompts. Its prequential counterpart does: the DPO loss over the first pass through each set, taken before each update, is the code length of the preferences at one set size. Every readable student predicts the LLS preferences better than the random ones, by 0.14 to 0.38 nats per pair (teacher 0.316 against 0.691; rnj-1 0.558 against 0.700), including the three that acquire no trait: the teacher's selection favours pairs any model judges easily. Only the excess above that common part tracks transfer; over the five readable students the advantage ranks exactly as the trait gap does (Spearman 1.0; Qwen2.5-7B's first pass was unstable and is not readable). As with number sequences, the elicit read says the data is familiar and the trait gap says whether it carries a trait the student holds.

4 Emergent misalignment

Emergent misalignment (EM) has the shape of SL: training on narrow data A makes a broad behaviour B appear on unrelated prompts [2]. We measured EM on Qwen2.5-14B-Instruct for 30 published datasets with the recipe of Turner et al. [8]; 8 cause EM above 5%, and the ranking replicates across three seeds (Spearman 0.98–0.99). Scores of A alone do not separate them; scores of B do. Let B^+ be misaligned responses to the standard EM prompts Q, taken from our fine-tuned runs (judged aligned < 30, coherent > 50), and B^- aligned responses from the same runs to the same prompts.

The misaligned persona is held. On Qwen2.5-14B-Instruct, $s_\theta^{\mathrm{PG}}(B^+) = 0.89$–$0.93$ with the model's own responses as $\Delta_{\mathrm{full}}^{\mathrm{PG}}$: elicit-side. All six models we tested (Qwen2.5 0.5B–14B, the 14B base model and OLMo-2-7B) score 0.56–1, and every one shows EM after fine-tuning (5–28%). EM therefore elicits a persona the model already represents.

Metric 3 (Distance to B, DB)**.**

$$\Delta_\theta^{\mathrm{DB}} \;=\; \frac{1}{|Q|}\sum_{q\in Q}\Big[\bar{L}_\theta\big(B_q^+\big) - \bar{L}_\theta\big(B_q^-\big)\Big],$$

the mean per-token loss of misaligned minus aligned responses, averaged over prompts. The predictor for a dataset A is $\Delta_{\theta_k(A)}^{\mathrm{DB}}$, read after $k = 40$ fine-tuning steps on A (640 examples), with every response from A's generator family removed from $B^{\pm}$.

Which data elicits it is predictable. Across the 30 datasets, $-\Delta_{\theta_{40}(A)}^{\mathrm{DB}}$ predicts the EM the full fine-tune reaches (Spearman +0.57; AUC 0.78 for EM above 5%), and +0.55 after partialling out the untouched model's loss on A. Within each generator family the order is nearly exact (Spearman +0.62 to +1.00). The direction of the first update alone, before any training, gives +0.64 (AUC 0.76; +0.34 beyond loss). On $B^{\pm}$ after 40 steps, the ratio $\Delta_\theta^{\mathrm{PG}}(B^+)/\Delta_\theta^{\mathrm{PG}}(B^-)$ rises above 1 for EM-causing data (Turner et al.'s three datasets 1.11–1.22, against 0.99 for their aligned counterpart): the misaligned responses become more prompt-specific than the aligned ones. Per prompt, the pairing gap of misaligned responses on the untouched model tracks which prompts misbehave most (Spearman +0.34 within models over the eight standard prompts). In elicit-versus-teach terms, the

score says the persona is held, and narrow versus emergent depends on whether fitting A recruits it. Training on responses the model believes are false increases EM [9]; on our model the on-policy version of their mathematics experiment reproduces the direction weakly (0%, 0%, 5% for correct, subtly wrong and absurd responses), and both our reads order the three sets correctly.

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